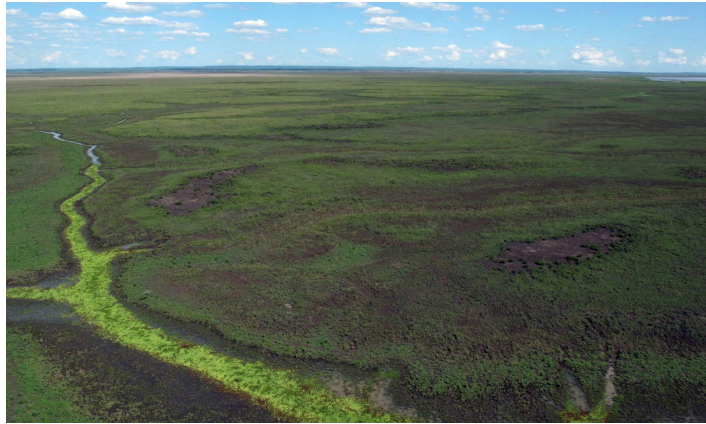


Mathematical Modelling of Natural Systems

Assignment 3: Machine Learning Methods to Retrieve Forest Biophysical Parameters

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1 General Concepts of Machine Learning

The model with the lowest root mean square error is model B as points in the model are closest to the line $y = x$ which indicates that *predicted* = *observed* - or in other words, it is a better model for predicting unseen results in the testing subset. RMSE is a good measure of how accurately the model predicts the result. Indeed we can see the equation for calculating RMSE is $RMSE = \sqrt{\frac{1}{N_t} \sum_{i=1}^{N_t} (y_i - \hat{y}_i)^2}$ is minimised when $|y_i - \hat{y}_i|$ is minimised for each i , which happens when y_i is close to $\hat{y}_i \iff$ the observed value is closer to the predicted value (or points lie closer to the red line).

The model with the lowest variance is model A, as the results are most clumped together, and there is the least spread of all the models. If we were to draw a region around the dots in this model, the area of the region would be less than the corresponding areas for the other 2 models. However, this model has high bias and high RMSE, and so is not a very accurate model.

The models with the lowest bias is model B, as lower bias means a better model. Indeed, using the bias-variance decomposition of the RMSE, $RMSE = \sqrt{bias^2 + variance}$ so it would make sense that low RMSE means low bias, so these must agree. Moreover, points do not favour any area in particular: in both models A and C the points are on the left half of the plot only, indicating bias towards that region. B does not have this issue.

2 Forest Biophysical Parameters

We want to compare the two ways of calculating aboveground biomass for a tree, found in equations (3.6) and (3.8).

Let $W1 = \frac{\pi}{4} d_i^2 h_i f_i \rho_i$ be the way to calculate aboveground biomass for a tree from (3.6), where d_i is in cm, h_i is in m and ρ_i is in gcm^{-3} .

Let $W2 = 0.0559(d_i^2 h_i \rho_i)$ be the way of calculating aboveground biomass for a tree from (3.8). This is given in Kg.

In order to compare these results, we must convert W1 into Kg, as at the moment it is in $cm^2 \cdot m \cdot g \cdot cm^{-3} = m \cdot g \cdot cm^{-1}$. To do this, we convert g to Kg, which scales W1 by a factor of $\frac{1}{1000}$, and cm^{-1} to m^{-1} , which scales W1 by a factor of 100. Hence the total conversion to Kg scales W1 by a factor of $\frac{100}{1000} = \frac{1}{10}$. Hence, letting $\bar{W1} = \frac{W1}{10}$ in Kg, the question asks us to find when $\bar{W1} > W2$.

$$\bar{W1} > W2 \iff \frac{1}{10} \cdot \frac{\pi}{4} d_i^2 h_i f_i \rho_i > 0.0559(d_i^2 h_i \rho_i) \iff f_i > \frac{40 \cdot 0.0559}{\pi} \approx 0.712(3.s.f).$$

So, in conclusion, when $f_i > 0.712$, the value of the tree aboveground biomass calculated in (3.6) is greater than the tree aboveground biomass calculated in the (3.8) way.

Dually, if $f_i \approx 0.712$ for a lot of trees in the forest we are looking at, then these two methods of calculating aboveground biomass can be used in conjunction with one another. If f_i is too different from 0.712, then the W2 method of calculating aboveground biomass is less accurate (as it relies on less physical features), so must be adapted in order to accurately encapsulate the aboveground biomass of the forest. To do this, we would have to find different α and β such that $W2 = \alpha(d_i^2 h_i \rho_i)^\beta$, that fits the biophysical parameters of the forest better.

Moreover, the forest form factor is defined as the “ratio of the volume of the tree trunk by the cylinder of the same height as the trunk with a sectional area equal to the sectional area of the trunk at breast height (1.3m)” [5]. If we assume the trunk to be conical as a simplification, then $f_i \approx 0.712 \implies$

$$\frac{Vol(Cylinder)}{Vol(Cone)} = 0.712 \implies \frac{\pi r_1^2 h}{\frac{\pi r_2^2 h}{3}} = 0.712 \implies 3 \frac{r_1^2}{r_2^2} = 0.712 \implies r_1 = \sqrt{\frac{0.712}{3}} r_2 \approx 0.487 r_2 \approx \frac{r_2}{2} (*),$$

where r_1 is the radius of the cylinder and r_2 is the radius of the cone. We can see what this looks like in fig.1. However, there are many other more complicated ways to calculate form factor that are more commonly used now [3].

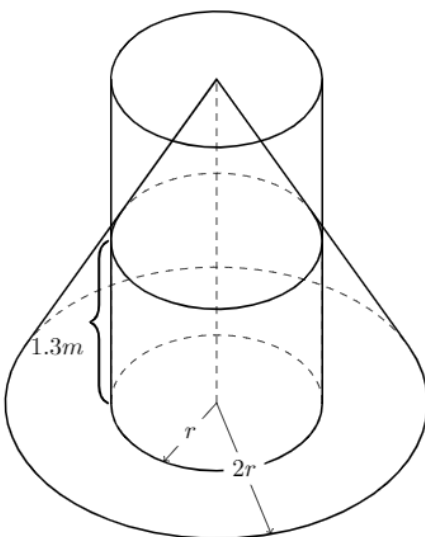


Figure 1: A visual representation of the ratio between the cone and cylinder, for trees with tree form factor around 0.712, by the calculations in (*). 1.3m is the breast height, in which the cone has the same radius as the cylinder.

3 Modelling Aboveground Biomass

3.1 We aim to show using the dataset `mz_biomass.cv` that less complex regression trees usually result in estimates of aboveground biomass with lower variance. We do this with the full tree and find that the variance is 6216. We then prune this full tree by increasing the parameter α in the sklearn library's `DecisionTreeRegressor` function, which corresponds to the α given in equation (1.9) in the notes. A higher value of α gives less leaves on the tree, and therefore it is less complex. The tree becomes least complex at $\alpha = 1090$, in which it becomes a single node. Increasing α from this value does not change the tree. With $\alpha = 50$, the tree is simplified to have only 17 leaf nodes, with a variance of 4213. With $\alpha = 150$, the tree only has 4 leaf nodes, with a variance of 2604, even less. We show these trees in fig.2.

We plot α against variance in fig.3 (i), and see a strong negative correlation, showing that less complex regression trees usually result in estimated aboveground biomass with lower variance, as required.

From this, we can calculate the value of α that give minimal variance. We get that the optimal α for reducing variance is $\alpha = 320$ (we can see this on fig.3 (i) as the minimum of the orange line), which yields a two leaved tree (fig.3 (ii)). This gives us the plot in fig.3 (iii), which shows how the model works against the testing subset.

We see that the regression model with the least variance predicts that the AGB is best predicted as one of 2 values. This could perhaps suggest the forests in Mozambique have 2 strong canopy layers. Indeed, Mozambique's forestry is made up of 67% semi-deciduous and 20% evergreen, 2 different kinds of trees[1], possibly leading to two main classes of aboveground biomass. However, we see from fig.3 (right) that it is not a very accurate model as there are a lot of points far away from the line $y = x$. Consequently, we could also explain these 2 values as the model under-fitting the data by making the model too simple, such that it does not describe the true variety of the forests.

A full tree, on the other, hand overfits the training dataset, and an overfitting gives high variance on the test subset [7]. This explains why having a less complex tree reduces the variance. Hence, we must search for better models. We do this in the next sections using ensemble methods.

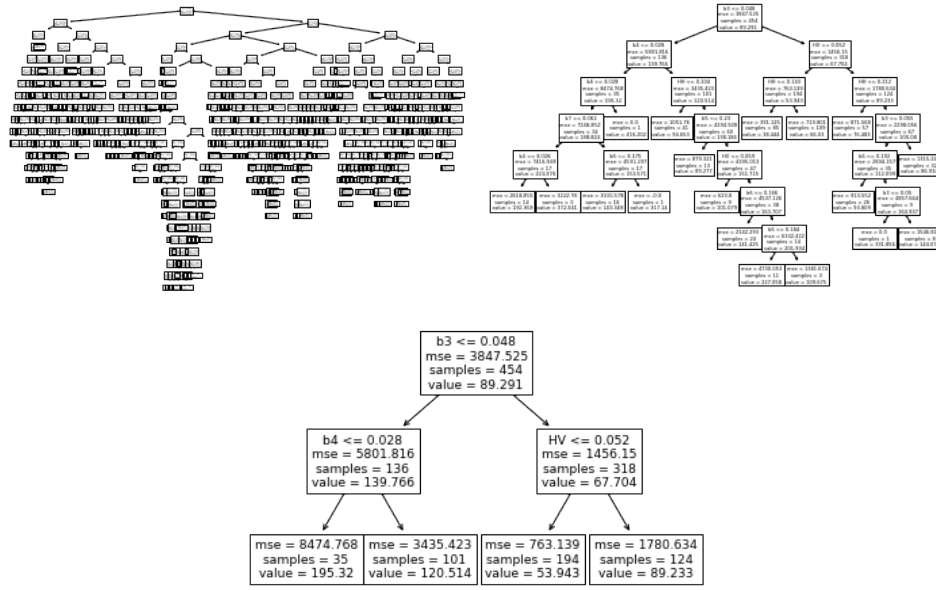


Figure 2: We show the full tree, and then the trees with $\alpha = 50$ and $\alpha = 150$. The variance reduces for less complex trees

3.2

In bagging, the tuning parameter we must test is the number of estimators. We search, as per the question, for the model with the lowest RMSE with estimators: $n \in \{100, 200, 300, 400, 500, 600, 700, 800, 900, 1000\}$. We find that the best model is given when $n = 600$.

For Boosting, the tuning parameters are the number of estimators and the learning rate of the model. We search for lowest RMSE in the 50 models given by number of estimators: $n \in \{100, 200, 300, \dots, 900, 1000\}$ and learning rate: $l \in \{0.001, 0.005, 0.01, 0.05, 0.1\}$. We find the best model to be with the parameters $(n, l) = (100, 0.005)$.

We tabulate the results of tuning the parameters using bagging and boosting in comparison to linear regression:

Comparing the fitting of the best algorithms				
Algorithm	RMSE (bias-variance)	RMSE (pre-dicted/observed)	Bias	Variance
Full Regression Tree	79.713	108.4	-11.751	6216
Bagging	53.374	57.1	-5.7684	2815.55
Boosting	53.2279	58.8	2.876	2824.98

From this we can see that both bagging and boosting are better models than the full regression tree, as the RMSE, bias and variance is lower for these models. This is unsurprising as ensemble methods, such as bagging and boosting, are more complicated than decision tree methods; they rely on the "wisdom of crowds" idea - run a lot of different models (in either sequence or parallel) and take in some way the best bits of these models for the final fit[7].

The model with the lowest RMSE is the boosting model. However, there really is not much difference between the RMSE's calculated in the bias-variance method between bagging and boosting (only 0.146) and so, in order to pick a best model, we must decide whether to prioritise smaller (absolute

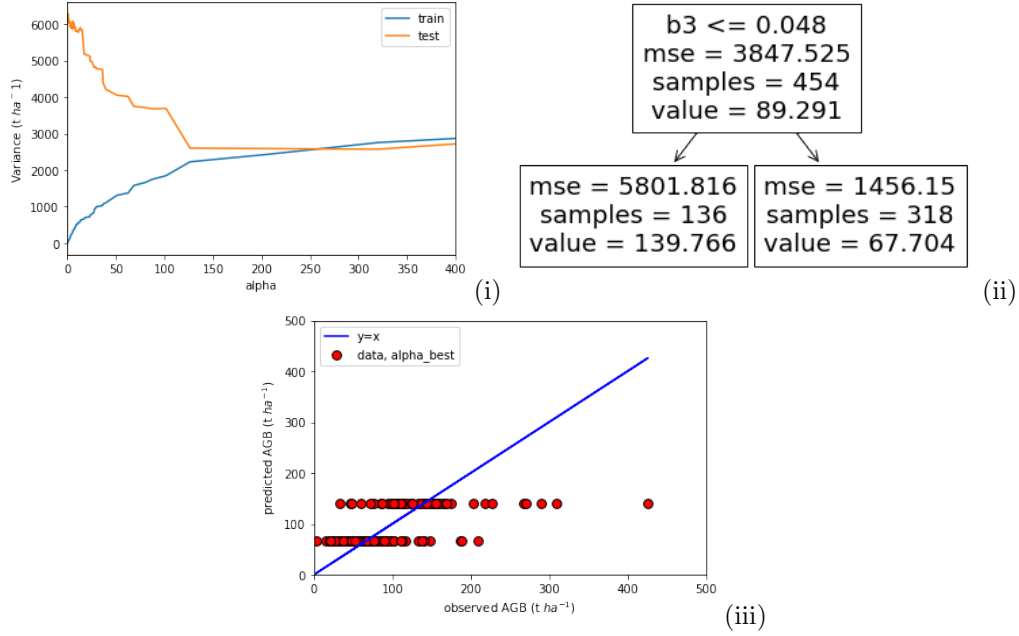


Figure 3: (i) We plot regression trees with different values of α against the variance for these trees against the testing data. (ii) shows the regression tree for $\alpha = 320$. (iii) plots the testing subset's real values of AGB against the values that the model predicts it should have, based on the predictors.

value of) bias or lower variance. If we want lower variance, then we should use the bagging model. If we want smaller bias, then we should use the boosting model. If we additionally compare the RMSE calculated using the formula $RMSE = \sqrt{\frac{1}{N_t} \sum_{i=1}^{N_t} (y_i - \hat{y}_i)^2}$, we see that bagging has lower error than Boosting. Therefore we can argue that the best model is the bagging model with 600 estimators.

3.3 In the Random Forest method, the tuning parameters are number of estimators (n), and maximum number of predictors at each node (m). We search for $n \in \{200, 400, 600, 800, 1000\}$ and for $m \in \{1, 2, 3, 4, 5, 6\}$.

Using the Gradient Boosting method, the tuning parameters are number of estimators (n), learning rate (l), maximum tree depth (d), and maximum number of predictors at each node (m). We search for $n \in \{200, 400, 600, 800, 1000\}$, $l \in \{0.001, 0.005, 0.01, 0.05, 0.1\}$. $d \in \{3, 5, 7, 9\}$ and for $m \in \{1, 2, 3, 4, 5, 6\}$.

We search for the best model, i.e. with the lowest RMSE.

i) We first use the bias-variance method of calculating the RMSE. We find that for Random Forests, the best model is given with parameters $(n, m) = (200, 1)$. For Gradient Boosting, we find the best model is $(n, l, d, m) = (200, 0.01, 2, 3)$ We tabulate the results for these models below:

Comparing the fitting of the best algorithms			
Algorithm	RMSE (bias-variance)	Bias	Variance
Full regression tree	79.713	-11.751	6216
Random Forest (i)	51.0529	-5.5729	2575.34
Gradient Boosting (i)	49.8824	-3.8314	2473.58

From this, we conclude that the best model is the Gradient Boosting model, which has lowest RMSE, lowest bias and lowest variance.

ii) Now we use the relationship between the observed and predicted values to calculate the RMSE using the formula $RMSE = \sqrt{\frac{1}{N_t} \sum_{i=1}^{N_t} (y_i - \hat{y}_i)^2}$, and rerun the process of finding the best model. For Random Forest, we have $(n, m) = (200, 6)$ and for gradient boosting in this case we have $(n, l, d, m) = (1000, 6, 0.5, 7)$. We label these models “Random Forest (ii)” and “Gradient Boosting (ii)” to distinguish these from the models calculated in **3.3 i)** (which we shall refer to as “Random Forest (i)” and “Gradient Boosting (i)”). We tabulate the results:

Comparing the fitting of the best algorithms (ii)			
Algorithm	RMSE	Bias	Variance
Full Regression Tree	79.713	-11.751	6216
Random Forest (ii)	56.599	-6.455	2885.42
Gradient Boosting (ii)	42.7373	-9.9995	3639.56

From this we can see that the model with the lowest RMSE is Gradient Boosting (ii), noticeably in this case. However, there is a trade off with higher bias and variance in this case, which is not the case when using the bias/variance method of calculating the RMSE as lowering this RMSE is linked with lowering bias and variance due to their linked nature. We might suggest, then that bias/variance is more useful as a measure of how good the model is, as a lower RMSE means a better model all round.

Conclusion We tabulate all our best models below and compare them:

Comparing the fitting of the best algorithms			
Algorithm	RMSE	Bias	Variance
Full regression tree	79.713	-11.751	6216
Bagging	53.374	-5.7684	2815.55
Boosting	53.2279	2.876	2824.98
Random Forest (i)	51.0529	-5.5729	2575.34
Gradient Boosting (i)	49.8824	-3.8314	2473.58
Random Forest (ii)	56.599	-6.455	2885.42
Gradient Boosting (ii)	42.7373	-9.9995	3639.56

From this we can conclude that the best model for our data is Gradient Boosting (i), with 200 estimators, learning rate 0.01, 2 max tree depth and a maximum of 3 predictors for each nodes. It is unsurprising that the Gradient Boosting and Random Forest models create a better fit than

with bagging or boosting as they are more complicated algorithms; the Random Forests method is designed to be similar to bagging, but to reduce the variance of it by making the sequence of models more independent with randomness (this is seen in comparing variance in Random Forests (i) with bagging); Gradient Boosting is designed so that it works similarly to boosting, but only working on the residuals, thus keeping what was correct in the earlier runs of the model still correct, and minimising the RMSE (in fact, it can be shown that AdaBoost is a specific kind of Gradient Boosting, when using the exponential loss function[7]). We see that both in cases (i) and (ii) that Gradient Boosting is better than the Random forest method. In [4], Ravanshad claims that “It has been shown that [Gradient Boosting] performs better than [Random Forests] if parameters [are] tuned carefully”. In our search for the best Gradient Boosting parameters, we used 600 different combinations of our tuning parameters in order to find the best performing model. This can help us understand why the Gradient boosting method performed better than the Random Forest method. However, it did take a lot longer to form this model, so for even bigger datasets, or for even more tuning parameter possibilities, the random forests method would be much more practical.

We compare the full regression tree with our best model in the graphs in fig.4. We can see that our Gradient Boosting (i) model is a much better fit, as points are closer to the line $y = x$, showing the lower RMSE and bias. We can also tell that the variance is much lower too.

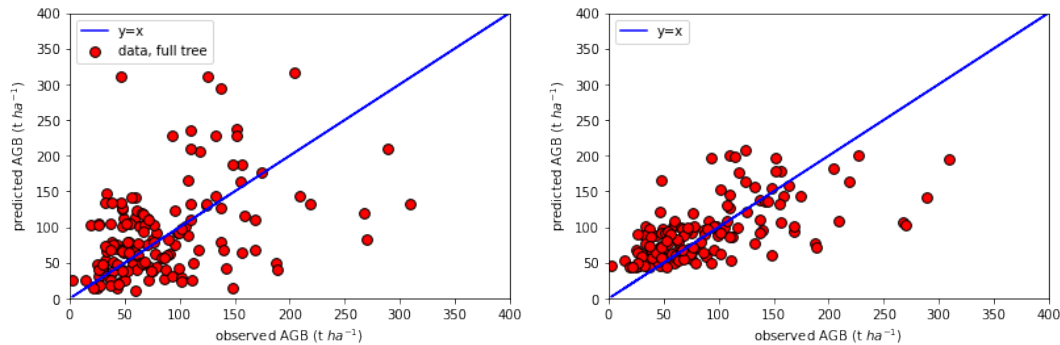


Figure 4: (left) is our base model: the full regression tree; (right) is our winning model, Gradient Boosting (i), with 200 estimators, learning rate 0.01, 2 max tree depth and a maximum of 3 predictors for each nodes.

Extending our search We can use the Bayesian Hyper-parameter tuning method to find an even better model, in even less time. We use this search and find the best model has 112 estimator, max tree depth of 2, a learning rate of 0.030583 and maximum predictors per node of 4. This gives an RMSE of 51.8054, a bias of -4.0345 and a variance of 2667.5195. These values are not much different, and are even worse than some of the models we have calculated. However, the code for this was much faster (especially than the Gradient Boosting code), and so is more practical for larger datasets, for example. Perhaps we cannot get much below the values of RMSE, bias and variance that we have already discovered. Indeed, there is a tradeoff between variance and bias, so that not both of them can be small. Moreover, there is noise in any dataset - the σ_ϵ^2 term in (1.17) of the notes: the irreducible error that is present in any model. Perhaps this noise prevents us from minimising the RMSE much more than we have already.

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